

MOOSEDev — State of the Evidence (consolidated evaluation, 2026-06-23)

A session-long, multi-instrument evaluation of one question: **does structured (typed-graph) project memory beat good free-text / vector memory for a coding agent?** Every instrument is reuse-first, pre-registers a strict verdict, grades against primary or source-derived ground truth, and is reproducible from immutable transcripts. The **source of truth is the project knowledge graph** (typed Lessons, linked by IRI); this file is the human-readable mirror (CLAUDE.md invariant #2). Private-corpus note: `trivyn` is private, only aggregate numbers appear; `rust-rfcs` and `codegraph` are public.

Executive summary

MOOSEDev's demonstrable value is **completeness/structure + currency** — **NOT retrieval precision**. Against a *real* competitor (mem0) on a neutral public corpus, structured memory answers **completeness, negation, and traversal** questions that vector/free-text memory **structurally cannot** (B2 1.00 vs mem0 ~0.15), while **tying** mem0 on its home turf (relevance recall, currency). It is a **superset**, not a sidegrade. The longitudinal "fights comprehension debt over months" thesis, the project's actual purpose, remains **unmeasured**.

dimension	instrument	result	status
Retrieval precision	hybrid A/B, scale-degradation, decomposition	constant offset, no compounding, saturates on clean data	≈ parity (3 nulls)
Capability (completeness / negation / traversal)	Stage 0–2 + real-competitor test vs mem0	B2 sweeps; mem0 ~0.15, real-docs-grep ~0.05; reproduced on 2 corpora	categorical win
Relevance + currency (vs mem0)	balanced competitor matrix	both arms call their tool reliably at ~35–40k tok; equal coverage	tie (superset)
Currency / anti-staleness	H4 reversal pairs	B2 100% current; free-text collapses on rank-inverted reversals	proven (rare-trigger)
Longitudinal (comprehension debt / months)	—	the actual thesis	untested

Verdict (see §9): don't kill it — *earn the thesis*. The evidence rules out the kill signal ("structured ≈ vector memory but costlier → redundant") and earns the expensive trial.

1. How to trust this (method)

The **arms ladder**, and how it *evolved* under criticism (each step a recorded correction):

- B0** cold (no memory) · **B1-md** rationale-as-markdown (agent grep) · **B1-rag** a `recall()` tool · **B2** the MOOSEDev typed graph (`get_relevant_context` / `query` / `sparql`).
- B1-rag/B1-md were content-parity** — the B2 graph *flattened to text* (AD `47f3f038`). A valid representation-only control, but "the graph wearing a markdown hat" (AD `b3205dcb`): it hands free-text MOOSEDev's capture+curation for free.

- So the headline baseline moved to **ecological / real competitors: B1-notes** (agent greps the project's real docs) and, this session, **B1-mem0** (the actual mem0 tool, capturing the raw docs its own way) — Lesson [440abc78](#) .

Discipline: pre-registered verdicts; [\\$0](#) probes that rerun from the stores with no agent; **regrade from immutable transcripts** after any grader change; a **strict LLM-judge** for cross-vocabulary fairness, *validated by reproducing the structured arm's score*; ground truth from **primary source / SPARQL-derived sets**, never an arm's own prose. **Corpora:** trivyn (private, ~416 records, 42 supersession chains), rust-rfcs (public, decision-scale), codegraph (public, neutral, not our doc style — the external-validity workhorse).

2. Retrieval precision — PARITY (three nulls)

Structured retrieval $\approx$ good BM25/hybrid. If you only need "find the relevant record," RAG is fine.

- **Hybrid dense@BM25F (AD [b933bd10](#))**: a *head-of-ranking re-ranker*, not a recall expander — recovers nothing BM25F misses within k=10 but promotes the current record (recall@3 +0.216 on hard old-framing) and spends **−19% agent tokens**. Accuracy near-ceiling for both arms. Lesson [410d6c73](#) .
- **Scale-degradation (rust-rfcs, N=50→634)**: B2 beats B1-rag at every N (hit@5 0.84 vs 0.60 at N=634) — a real constant offset — but the gap does **not** widen: slope $\beta = -0.033$ [95% CI $-0.124, +0.059$], spans 0. **The "edge compounds with scale" thesis is KILLED**. Lesson [7b41be4f](#) .
- **Decomposition (split blob → typed nodes)**: clean corpora are retrieval-saturated (blob hit@5 = 1.00), so decomposition has no gap to fill. Lesson [8f06f329](#) .

Read: retrieval precision is not the differentiator, and does not justify the project on its own.

3. Capability — the categorical win (the core result)

A **possible-vs-impossible** axis, not a shared accuracy axis (Lesson [8a06ad1d](#)): *enumerate a complete typed set, find an absence, traverse a supersession/multi-hop link*. Top-k retrieval cannot do these at any k< all; a typed graph returns them in one query.

Ladder of evidence:

- **Stage 0 — tool ceiling (\$0, trivyn)**: for answer sets ≥ 16 , negations, and traversals, B1 BM25 recall@5 ≈ 0.00 –0.11 (needle in a 416-record haystack); reaching completeness needs ingesting the whole ~63K-token corpus; B2 returns the exact set in one SPARQL. Lesson [392855d0](#) .
- **Stage 1 — agentic (trivyn, N=3)**: pooled hard-class **B2 F1 0.94 / recall ~0.99** vs B1-rag 0.25 vs B0 ~0. Lesson [6a70b02e](#) .
- **Stage 2 — neutral public corpus (codegraph, N=2)**: reproduces — pooled **B2 0.89** vs B1-rag 0.34 vs B0 0, on someone else's docs with a richer graph than ours. Lesson [243c6c89](#) .

The definitive test — vs a REAL competitor (this session, codegraph). Replaces the graph-derived baseline with the *actual* tools, captured **whole-system**: B1-mem0 = mem0 ingesting the raw docs its own way (553 memories, verified faithful); B1-notes = the agent grepping the real shipped docs. Graded by a **B2-validated LLM judge** (credits paraphrase, rejects same-topic-not-same-rule):

class	B2	B1-mem0 (fair)	B1-notes (fair)
set-completeness	1.00	0.15	0.02
negation	1.00	0.17	0.01
supersession	1.00	0.15	0.15

mem0 surfaces ~15% of a large set (its top-k slice) and **never the whole** — and it's **capture-clean** (mem0 *holds* the 553 facts; it just can't enumerate/negate/traverse them). This is the strongest external-validity result we have: the categorical gap survives a real tool, on a public corpus, fairly graded.

4. The mem0 comparison — the balanced matrix

Testing only B2's turf would be a strawman, so the matrix also tests **mem0's home turf** (relevance recall + currency), with mem0 at full multi-signal strength (semantic + BM25 + entity):

dimension	B2	mem0	verdict
completeness / negation / supersession	1.00 (via <code>sparql</code> , ~78–167k tok)	~0.15	B2 categorical
simple relevance	cov 0.82 (<code>get_relevant_context</code> 4/4, ~35k tok)	cov 0.67–0.90 (~40k tok)	tie
currency (real CHANGELOG reversal)	1.00	1.00	tie (cold 0.00)

Synthesis — inverse trades. mem0 = cheap, foolproof, small-context; MOOSEDev = complete, structured, auditable. They **tie where mem0 is built to win**; MOOSEDev is a **strict superset on the structural axis**, which top-k cannot reach. *"Different tools for different jobs — one strictly broader on completeness/structure."*

(Multi-hop omitted from the table: size-1 expected sets make recall-only coverage degenerate; needs a precision-aware metric.)

5. Currency — proven (rare-triggering)

When a decision is later reversed, does memory serve the **current** answer or the **stale** one? On four real reversal pairs (×4 models, ×2 delivery regimes), **B2 is 100% current (25/25)**. Free-text is current on 3 of 4 pairs but **collapses on the rank-inverted pair** (`glrc` , where the superseded record out-ranks the current one in BM25): in the push/oracle regime B1-rag serves the stale answer **1/13 current (8%)** vs B2 **13/13**. Structured current-only retrieval is a **guarantee**; free-text currency is a content/rank/regime-dependent **gamble**. Real, but rare-triggering — rich capture suppresses how often reversals bite. Rationale `51338ce8` , `abe11031` ; Lesson `9fc21dc0` (grading). Full detail: `bench/RESULTS.md` .

6. Efficiency

Mixed / parity — **accuracy is the win, not tokens**. Hybrid retrieval saves ~19% agent tokens on the currency corpus. MOOSEDev's *read* cost is moderate and **query-mechanism-dependent**, not fixed: targeted `get_relevant_context` (~35k) ≈ mem0's `recall` ; completeness via SPARQL is ~78–167k for a 122-set agentic run — a capability mem0 lacks at any cost. (An alarming "1.68M token" figure turned out to be a harness bug, §7.)

7. Honesty trail — why the numbers are trustworthy

The evaluation repeatedly caught and corrected its *own* errors; that auditability is the point.

- A false +20pp "drift" win was overturned by a free regrade over immutable transcripts — a grading artifact (a stale - substring flag + a token-balloon "escape" proxy). A grader can *manufacture* a signal, not only hide one. Lesson `a03dbe13` .
- The LLM judge caught its own format bug — it scored B2 0.00 on tasks B2 strict-scored 1.00 (B2 formats sets as fenced "title | IRI" blocks a bullet-extractor misses). The B2-validation check flagged it; the judge now reads the full answer. Lesson `a6529240` .
- A dead-MCP bug nearly produced a wrong conclusion. A `.env` loader I added clobbered `MOOSEDEV_BIN` with an unexpanded `$R00T` , so the moosedev MCP never launched and every B2 agent fell back to grepping the store (0.5–1.2M tokens). This masqueraded as "mem0 wins relevance" / "B2 thrashes" / a weak-model tool-calling gap. With the bin fixed, gpt-5.4-mini calls `get_relevant_context` 4/4 at ~35k tokens and B2 ties mem0 on relevance. The fix also retired the "agent query-competence" story. Lesson `0a890ea9` (supersedes the turf and efficiency claims → `e80db883` , `4d34f8c6`).

The meta-point: a plausible-but-wrong conclusion almost stuck, and an auditable, re-checkable process caught it — which is precisely the failure mode (confident, fluent, wrong) the whole project is betting it can reduce.

8. Honest scope — what is NOT shown

- **Governance / auditability** (per-claim provenance, supersede-not-overwrite, validation) is MOOSEDev's deepest potential edge over mem0 — but it's **by design, unmeasured**.
- **Comprehension debt over months** — the *actual reason the project exists* — has **zero evidence**. The capability win is the point-in-time retrieval slice, not the longitudinal payoff.
- **Conditional on structured capture**. The win requires you to *capture* typed knowledge — friction mem0 doesn't have. On a flat capture, B2 degrades to free-text parity (Lesson `0b2a0f80`).
- **Single agent family** (codex / gpt-5.x); relevance markers were prior-contaminated (cold B0 0.33–0.80), so the relevance *tie* rests on tool-usage + token parity, not coverage — cleaner project-specific relevance tasks are a follow-up.

9. Verdict — don't kill it, earn the thesis

The kill signal would have been *structured memory* $\approx$ *vector/free-text memory on what matters, but costlier* → *redundant*. We looked for it against a real competitor and found the **opposite**: a categorical, reproduced, capture-clean capability (completeness / negation / traversal) that mem0 **cannot** match, with **parity** where mem0 is strong — i.e. a superset. That clears the bar to keep investing.

It does **not** prove the big claims (governance, comprehension-debt-over-months); those stay hypotheses. So the benchmark did its job: it ruled out "redundant RAG" and **earned the months-long in-anger trial** (AD `d3d4d7b1`) — which is the real test of the thesis, to be instrumented up front (pre-registered "recover the why" probes, a firing tally, a kill condition). Killing MOOSEDev now would be killing it right before the test that matters, on evidence that already shows a real, alternative-beating capability.

10. Artifacts & reproduce

- **Graph (source of truth)**: capability — Lessons `392855d0` , `6a70b02e` , `243c6c89` , `8a06ad1d` ; competitor method — `440abc78` ; grading — `a6529240` , `a03dbe13` , `9fc21dc0` ; the dead-MCP correction — `0a890ea9` , `e80db883` , `4d34f8c6` ; retrieval nulls — `7b41be4f` , `8f06f329` , `410d6c73` . ADs `47f3f038` (bench), `b3205dcb` (ecological arms), `b933bd10` (hybrid), `d3d4d7b1` (in-anger trial). Query via the moosedev MCP.
- **Harness (bench/ , regrade-safe)**: retrieval — `hybrid_*` , `scale_*` , `run_hybrid_ab.sh` , `run_scale_degradation.sh` ; capability — `capability_build.py` , `grade_set.py` , `run_capability.sh` ; competitor — `mem0_build.py` (mem0 ingests

raw docs → on-disk qdrant), `mem0_mcp/server.py`, `regrade_judge.py` (B2-validated LLM judge); config — `config.py` .
Currency detail: `RESULTS.md` .

- **Reproduce:** `$0` probes rerun from the stores with no agent; agentic matrices regrade from logged transcripts; competitor build needs `OPENROUTER_API_KEY` in `.env` , then `mem0_build.py --corpus codegraph --reset` and `CORPUS=codegraph ARMS="B2 B1-notes B1-mem0" ./run_capability.sh` .